

2.5 Concept of Data Structure

ڈیٹا سٹرکچر ڈیٹا کو منظم طریقے سے رکھنے کا نظام ہے تاکہ الگورتھم اسے جلدی استعمال کر سکے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 11 of 16

★ Golden

2.5 Concept of Data Structure ★ Golden

LEARN IT

A **data structure** is a systematic way of **organising, storing and managing** data so that an algorithm can access, modify and process it efficiently.

Real-life analogy. Books on a shelf are arranged in a set order, so you can find a title quickly. Files in folders are grouped on purpose. Data structures do the same job inside computer memory.

1. Linear data structures

Elements stand in a strict sequence, one after another. Each element (except the first and last) has exactly **one predecessor** and **one successor**.

Examples: Array, Linked List, Stack, Queue.

Linear (sequence)



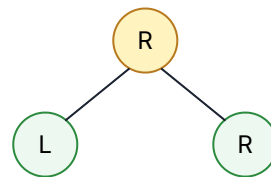
one after another

2. Non-linear data structures

Elements are connected in a hierarchy or a network. One element can connect to **several** others at the same time.

Examples: Tree, Graph.

Non-linear (branch / net)



Linear: a line of neighbours. Non-linear: a node with more than one connection.

WORKED EXAMPLE

Worked example: classify each everyday picture.

1. Students standing in a single assembly line → **linear** (each has one in front and one behind).
2. A family tree of grandparents, parents and children → **non-linear** (one parent, several children).
3. Roads linking Karachi, Hyderabad, Sukkur and Larkana in many ways → **non-linear** (a network).

REMEMBER IT

Ask: "Does each item have at most one next item?" Yes → linear. "Can one item link to many, or can links form a net?" → non-linear.

COMMON MISTAKE

Mistake: calling a tree "linear" because you can walk it from top to bottom.

Fix: walking is traversal. The **shape** is hierarchical: one parent, many children. That is non-linear.

اردو میں سمجھیں

ڈیٹا سٹرکچر ڈیٹا کو منظم طریقے سے رکھنے کا نظام ہے تاکہ الگورتھم اسے جلدی استعمال کر سکے۔ لکیری ڈھانچوں میں عناصر قطار میں ہوتے ہیں (ایرے، لنکڈ لسٹ، سٹیک، کیو)۔ غیر لکیری میں ایک عنصر کئی سے جڑ سکتا ہے (ٹری، گراف)۔ الماری کی کتابیں اور فولڈرز کی مثال یاد رکھیں۔

EXAM TIP

Define data structure in full (organise, store, manage; efficient access and processing). Then split linear vs non-linear with the "one predecessor and one successor" test, and list the four linear and two non-linear examples. The shelf / folder analogy earns a mark. MCQs: "Which is non-linear?" → Tree or Graph.

CHECK YOURSELF

1. Define data structure.
2. Name four linear and two non-linear examples.
3. What is true of each middle element in a linear structure?

Answers: 1. A systematic way of organising, storing and managing data for efficient access, modification and processing. 2. Linear: array, linked list, stack, queue. Non-linear: tree, graph. 3. It has exactly one predecessor and one successor.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A data structure is:

- a. An organised way to store and access data
- b. A random pile
- c. A font
- d. A sprint

MCQ 2. We pick a structure for:

- a. The operations we need (fast insert? fast search?)
- b. The logo
- c. The teacher's birthday
- d. The PDF theme

MCQ 3. Linear vs non-linear is about:

- a. Whether elements form a sequence vs branches/links
- b. Colour
- c. File format .py vs .txt
- d. Agile vs waterfall

MCQ 4. An array gives fast:

- a. Index access $O(1)$
- b. Insert at front $O(1)$ always
- c. Arbitrary graph neighbours
- d. Sorted insert $O(1)$ always

MCQ 5. This is ★ because papers ask:

- a. Define + why we need DS + one example
- b. Only careers
- c. Only Figma
- d. Only OSI

MCQ 6. A queue of print jobs is a DS chosen because:

- a. Order of arrival matters (FIFO)
- b. LIFO undo
- c. Hierarchy of boss-staff
- d. Random friends links

Extra short questions

1. Define data structure.
2. Why not always use one big list?
3. Give one linear and one non-linear example.
4. What is an operation on a DS?

Extra long question

1. "If you pick the wrong structure, the algorithm looks harder." Explain with college examples of array, stack, queue, tree. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. A systematic way to organise data for efficient access and modification.
2. Some problems need LIFO, FIFO, hierarchy, or many-to-many links.
3. Linear: array/stack/queue. Non-linear: tree/graph.
4. Insert, delete, search, traverse, peek...

Long

1. Roll list: array. Undo typing: stack. Printer/canteen line: queue. Org chart/folders: tree.
Wrong pick: using a stack for a fair queue angers people.

Next lecture: 2.5.1 Linear Data Structures (Array, Linked List, Stack, Queue)